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(See 6.8)

MILITARY SPECIFICATION

INTERCOMMUNICATION STATIONS (HANDS-FREE OPERATED REPLY REPRODUCER) NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers LS-458/SIC "hands-free" operated intercommunication stations for use in Naval shipboard announcing systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power and High Power Pulse), General Specification for.
MIL-C-62 - Capacitors, Fixed, Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized, General Specification for.
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte) Tantalum, General Specification for.
MIL-L-12606 - Loudspeaker, Permanent Magnet (Unencased, 3-inch and 4-inch Diameter Cone, 2-Watt; Fungus- and Immersion-Resistant), Type LS-445/U, M12606-01 and M12606.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, Provisioned Hem and (Repair Parts); Packaging of.
MIL-I-24078 - Intercommunication Stations, Naval Shipboard.
MIL-C-24643 - Cable and Cord, Electrical, Low Smoke, for Shipboard Use, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-454 - Electronic Equipment Standard General Requirements for.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-2036 - General Requirements for Electronic Equipment Specifications.

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

MILITARY

9000-S6502-74117 - Receptacle Announcing Equipment Plug and Receptacle Sym. 723, 724, and 725.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Code 280.6, Portsmouth, NH 03801.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Description. The LS-458/SIC intercommunication station, hereinafter referred to as the "equipment", shall have selector switches for calling two stations, a busy light, one press to-talk switch, a microphone receptacle, one speaker-microphone switch, one remote normal switch and one volume control switch. The equipment shall have a transistorized amplifier-speaker for voice amplification and transmission and shall have a signal lamp associated with each station selector switch. The equipment shall be installed in protected locations.

3.3 General features. The equipment shall be in accordance with the appropriate sections of MIL-STD-2036 in addition to the requirements specified herein.

3.4 Materials. Materials used in the construction of the equipment shall be as specified herein or of an equivalent material when approved by the command or agency concerned (see 3.3).

3.5 Design. The equipment shall provide naturalness and intelligibility of speech under all conditions, as described in ASTM F 1166. Accessibility for servicing shall be in accordance with the requirements of ASTM F 1166.

3.6 Fuseholders. Fuseholders shall be of a spring clip type and shall be silver plated.

3.7 Switches. The handles or keys of rotary type selector switches shall be provided with mechanical stops to prevent undue strain on the switch handle and shaft, and to prevent rotation of the fixed portion of the switch. All switches shall, after 100,000 operations, when energized at their rated loads, perform both mechanical and electrical functions for which they are intended.

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3.8 Interchangeability. The equipment shall conform to the following:

(a) The size and weight shall be greater than the following:

Case	Overall (including mounting plate and handle grips)
Width: 165 millimeters (mm)	165 mm
Depth 203 mm	229 mm
Height 216 mm	248 mm
Weight 11.34 kilograms (kg)	

(b) The microphone receptacle shall be in accordance with 9000-S6502-74117. Wiring to this receptacle shall be as shown in MIL-P-24078.

3.9 Enclosure. A complete equipment enclosure shall consist of a case and removable front panel designed for mounting on a bulkhead and in a panel. The enclosure shall be steel or aluminum. The enclosure shall be drip-proof in accordance with MIL-STD-2036.

3.10 Front panel and case securing bolts. Through bolting for the purpose of securing the front panel to the case and for securing the case in its mounted position will be permitted. All securing bolts shall have hexagonal heads with screwdriver slots. Sufficient space shall be allowed around bolt head surfaces to permit removal with a socket wrench.

3.11 Front panel surface. The front panel shall be free of unnecessary projections. The depth of all operating controls in the nonoperated position shall be approximately the same distance from the front panel surface.

3.12 Handle grips. Handle grips shall be provided on the front panel to facilitate withdrawal of the front panel from the case.

3.13 Electrical connections.

3.13.1 Ship's wiring connections. A terminal board shall be provided for connecting ship's wiring to the equipment. The terminal board location shall be such that ship's wiring can be connected to the terminal board without removing the equipment from its case. Sufficient terminals shall be provided on the terminal board to allow ship's wiring connections for all modes of operation.

3.13.2 Plug-in connectors. Multi-contact plug and jack connectors shall be provided which will disconnect the unit from ship's wiring when it is removed from its case. The plug shall be mounted on the equipment and the jack shall be mounted in the rear of the case such that the plug engages in the jack when the equipment is placed in its case. The connectors shall employ a floating type of mount to assure positive alignment and positive connection.

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3.13.2.1 Alignment guides. Alignment guides shall be provided to assure positive alignment between the plug and jack connectors. The guides shall support the equipment during insertion and removal intervals while in the case. The guides shall prevent any damage to the plug-in connectors or any part of the equipment during insertion and removal of the unit from its case.

3.14 Drainage. Drainage shall be provided for any panel recesses which otherwise would be susceptible to collection of water.

3.15 Control knobs. The size and shape of the operating control knobs shall be approved by the procuring activity for the function for which they are intended.

3.15.1 Switch controls. The position of all switch controls shall be clearly visible in the normal or operated position.

3.15.2 Knobs and shafts. Knobs and shafts shall have flat or rectangular shapes on the connecting parts and shall be secured with a through roll-pin or other means approved by the procuring activity.

3.16 Gaskets. Gaskets shall be used, if necessary, for making drip-proof seals between the case and front panel assembly. Gaskets shall be used for operating controls of the translating and rotary motion types.

3.17 Servicing. With the equipment removed from its case, all components shall be readily accessible for ease of replacement. When the equipment is removed from its case for servicing, all component parts shall be protected from possible damage when placed on any of its sides.

3.18 Transformers. Transformers shall be grade 5, class S, life expectancy X, as specified in MIL-T-27. Where special shielding problems exist, such as in input circuits of audio amplifiers, the use of encased type transformers employing high permeability alloy shields will be permitted.

3.19 Relays. Relays, if used, shall conform to Requirement 57 of MIL-STD-454 and shall have dustproof covers.

3.20 Illumination. All illumination shall be red in accordance with MIL-STD-2036. Illumination lamps shall be located to permit direct replacement of each lamp individually from the exterior of the watertight enclosure.

3.21 Signal indicator lamps. The signal indicator lamps shall be individually located adjacent to the associated station selector switches. Signal indicator lamps shall be subminiature, omnidirectional, red plastic dome using a type T-1-3/4 lamp.

3.22 Station identification. The station identification shall be a durable, abrasive-resistant laminated plastic strip suitable for engraving. The laminations shall be opaque black and translucent white. Guide lines shall be engraved on the marker to indicate the limits of the area to be engraved with the station designation. The engraved areas shall be at a sufficient depth to expose the translucent white lamination. The marker shall be a single strip removable from the front of the equipment.

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3.23 Rectifiers. Silicon type rectifiers shall be used to supply high voltage for the amplifier. A resistance shall be inserted in the high voltage circuit as a surge current limiter. Rectifiers shall be selected from approved types listed in MIL-STD-701 and MIL-M-38510.

3.24 Operational features. The operational features of the system shall be as specified in 3.24.1 and 3.24.10, inclusive. (Master stations referred to are type LS-519/SIC or LS-518/SIC units manufactured in accordance with MIL-I-24078.)

3.24.1 LS-458/SIC calling master station. Operate station selector switch. If station selected is not busy (indicated by busy lamp) operate talk-listen switch and talk into speaker.

3.24.2 Busy master station. If the master station is busy, the busy indicator lamp will light. This is an indication that the master station should not be called.

3.24.3 Operation of equipment with remote reproducer. This system shall consist of one LS-458/SIC station, one remote reproducer with one normally closed single pole switch. Circuits shall be provided in the equipment for the following operations:

(a) Operation of remote reproducer switch will connect remote reproducer to the input of the equipment amplifier and connect the equipment reproducer to the output of the equipment amplifier. The remote reproducer can then transmit speech.
(b) Operation of station selector switch on the equipment will connect the remote reproducer to input of the equipment amplifier and connect the equipment reproducer to output of the equipment amplifier. Speech can then be transmitted from the remote reproducer without operation of the remote reproducer switch.
(c) With equipment selector switch operated, this station can transmit speech to remote reproducer by operation of press-to-talk switch. Remote reproducer can reply with hands-free operation when the equipment "press to talk" switch is in restored position.

3.24.4 Master station calling equipment station. The equipment shall operate as follows:

(a) Normal operation to answer a call. Press talk-listen switch and talk into speaker. (However, even though unnecessary, if the conventional method of selecting the calling station by pressing the station selector switch is used, a busy light at the answering station shall light, but this shall not prevent the answering station from talking back to the calling station).
(b) Hands free operation. With "Remote-Normal" switch on LS-458/SIC station in "Remote" position, operation of station selector switch on master station shall put the equipment in "hands-free talk" position. Operation of press-to-talk switch on master station shall cause equipment station to be in "Listen" position. When master station depresses release switch both stations shall go to standby condition.

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3.24.5 Station selector switch. Restoring station selector switch to normal shall also turn off busy light if selected station is busy, and restore the station to the normal receiving position.

3.24.6 Operation of a station selector switch. Operation of a station selector switch shall cause the selected station (if not already busy) to become busy to all other stations in a system. Operation of this switch shall also cause the selecting station to become busy to all other stations in the system. Operation of a station selector switch shall energize the call light at the station selected provided the station is not busy.

3.24.7 Communication. Once communication has been established by operation of station selector switches, no other nonselected equipment shall be able to listen or to break in and talk to any units engaged in conversation.

3.24.8 Operation with handset or microphone. Operation with handset or microphone shall be identical to the normal operation, except that talk-listen functions will be performed remotely by the press-to-talk switch in the handset or microphone.

3.24.9 Operation under busy condition. If, on selection of either station selector switch a busy indication is received, the switch shall be released immediately. This will place the unit in condition for the reception of an incoming call. When a station has received a busy indication, it shall not be possible to hear speech from the busy station.

3.24.10 Parallel operation of units. When two equipments are connected in parallel operation, they shall be the same as for normal single operation with the following exceptions:

(a) Incoming speech shall be heard at both equipment.
(b) Either station may answer a call but not both. (Inadvertent answering by both units simultaneously will not cut off transmission of speech to the calling equipment.)
(c) Operation of a station selector at either of the paralleled equipments in addition to lighting the call light at the called station shall cause the call lamp in the unit to which it is paralleled to light.

3.25 Electrical features.

3.25.1 Circuit design. The circuit design shall perform the functions specified in 3.24 through 3.24.10.

3.25.2 Terminal markings. Terminal markings shall be in accordance with MIL-I-24078.

3.25.3 Interconnecting cables. Interconnecting cables shall conform to MIL-C-24643.

3.25.4 Volume control. The volume control shall provide a maximum attenuation of 30 ± 1 decibels (dB) in five steps of 6 ± 1 dB. No "OFF" position shall be incorporated in the control.

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3.25.5 Press-to-talk switch. The press-to-talk switch shall be a non-locking spring return switch. The nonoperated position to the switch shall be the normal incoming call position.

3.25.6 Speaker-microphone switch. The speaker-microphone switch shall be a DPDT toggle switch.

3.25.7 Speaker remote-normal switch. The speaker remote-normal switch shall be a SPST toggle switch.

3.25.8 Microphone and telephone handset. A connector shall be provided on all equipment. The purpose of the connector is to permit the use of either a microphone or telephone handset. The contractor of the intercommunication stations shall be responsible for the proper operation of the intercommunication stations when used with a type M-132/SIC microphone or an IC/MPH-2 handset. Facilities shall be provided to permit the use of a separate handset or microphone of the types specified herein.

3.25.9 Loudspeakers. Loudspeakers shall be type LS-445/U in accordance with MIL-L-12606 and shall be identical for all stations.

3.25.10 Transformer polarities. Identical instantaneous alternating current (ac) polarities for all transformers shall be obtained at physically identical transformer terminals on all stations, when all stations are energized from the same source.

3.26 Amplifier.

3.26.1 Transistors and metallic rectifier. Solid state devices shall be used in the design of all amplifiers. Solid state devices and power rectifiers shall be types chosen from approved types listed in MIL-STD-701 and MIL-M-38510.

3.26.2 Printed wiring. Printed wiring if used, shall be in accordance with MIL-STD-275 and type C lead terminations shall be employed.

3.26.2.1 Printed circuit board. Amplfier and power supply circuit boards shall be plug-in connecterized cards.

3.26.3 Protection of signal voltage winding of power transformer. Fuses shall be incorporated in the signal voltage circuit to prevent damage to the windings of the transformer due to defective signal voltage circuits.

3.26.3.1 Fuses. Fuses shall be inserted in each lead of the primary power supply in the unit.

3.26.4 Capacitors. Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in the amplifier assemblies.

3.26.4.1 Electrolytic capacitors. Electrolytic capacitors in accordance with MIL-C-62 shall be used in power supply filter circuits only.

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3.26.5 Personnel protection. In the case of voltages greater than 240 volts, a red warning plate shall be mounted in a conspicuous location in accordance with MIL-STD-2036.

3.26.6 Heat dissipation. Components which generate heat shall be mounted to allow maximum heat dissipation through the external case.

3.26.7 Rated power output. Rated power output shall be 3 watts at a maximum distortion of 10 percent.

3.26.8 Output voltage. The output voltage when delivering rated power shall be 70 volts ± 4 volts root mean square (rms).

3.26.9 Amplifier gain. The amplifier gain shall deliver rated output power at rated output voltage with a total harmonic distortion of 5 percent with a 1000 hertz (Hz) sine wave voltage equivalent in value to the voltage generated by the loudspeaker, when used as a microphone under the test conditions specified in 4.6.2 applied to the input terminals of the amplifier. The amplifier gain shall be such that the amplifier will also deliver rated output power at rated output voltage when tested with a type M-132/SIC microphone or equivalent which will deliver 8.7 millivolt (mV) into a resistive load of 150 ohms.

3.26.10 Output voltage regulation. With a steady state sine wave input voltage, of the same value used to determine amplifier gain, applied to the input terminals of the amplifier, a variation of the amplifiers output load from full load to no load shall not cause the output voltage to be greater than 99 volts rms at any frequency between 200 and 800 Hz.

3.26.11 Stability. The amplifier shall show no evidence of instability of oscillation under any condition of operation.

3.26.12 Amplifier noise and microphonics. The amplifier noise and microphonics shall be not greater than the following values when tested as specified in 4.6.7.

(a) Component stationary. Maximum measured voltage shall be not greater than 0.22 volts rms (50 dB below rated output voltage of 70 volts).
(b) Component vibrating. Maximum measured voltage shall be not greater than 0.31 volts rms (47 dB below rated output voltage of 70 volts).

3.26.13 Overall frequency response. The overall frequency response using the loudspeaker as a microphone shall fall within the limits of the overall frequency response curves shown on figure 1 when tested as specified in 4.6.4 and 4.6.5.

3.26.14 Minimum sound pressure. The loudspeaker, when mounted in the front panel, shall produce the following minimum sound pressure at a distance of 2 feet in front of and on axis with the loudspeaker when measured as specified in 4.6.3:

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	200 to 320	320 to 500
	85	95
Warble band Hz	500 to 800	800 to 1250
Output pressure dB 1/	99	99
	1250 to 2000	2000 to 3200
	99	99
	3200 to 5000	
	99	

1/ Above 0.000200 dynes per square centimeter (cm^2).

3.26.15 Minimum off-axis sound pressure. The loudspeaker, when mounted in the front panel, shall produce the following minimum 45-degree off-axis sound pressure at a distance of 2 feet when measured as specified in 4.6.3.

	200 to 300	300 to 500
	81	91
Warble band Hz	500 to 800	800 to 1250
Output pressure dB 1/	96	96
	1250 to 2000	2000 to 3200
	96	96
	3200 to 5000	
	91	

1/ Above 0.000200 dynes per cm^2.

3.26.16 Amplifier endurance. After being tested as specified in 4.6.6, the equipment shall operate normally in every respect. There shall be no visible damage to any electrical or mechanical component of the equipment.

3.26.17 Signaling circuits. The busy, and call signals shall not fail to indicate nor shall speech transmission be audible at a station receiving a busy signal when tested as specified in 4.6.8.

3.26.18 Amplifier output circuits. The paralleling of amplifier output circuits which can occur if two stations call a third station simultaneously shall not cause a loss of signal greater than 2 dB, nor shall the distortion exceed the maximum allowable distortion.

3.27 Designation and marking. An instruction plate containing abridged operating instructions shall be provided on the outside of the enclosure. Schematic circuit diagrams shall be provided and mounted inside the enclosure for use in servicing the equipment. The circuit diagram plates shall be of aluminum or plastic material. Designation plates shall be provided for all control knobs. These plates shall be located adjacent to the knobs and shall clearly indicate the function and direction of rotation of the knobs.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table I, which shall be performed in general in the order listed.

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TABLE I. First article inspection.

Inspection	Requirement	Inspection reference
Examination	3.3	MIL-STD-2036
Operating	3.24	4.6.1
Supply line voltage and frequency variation	3.3	MIL-STD-2036
Shielding and radio frequency noise reduction	3.3	MIL-STD-2036
Temperature and humidity	3.3	MIL-STD-2036
Insulation resistance	3.3	MIL-STD-2036
Amplifier gain	3.26.9	4.6.2
Amplifier noise and microphonics	3.26.12	4.6.7
Overall frequency response	3.26.13	4.6.4 and 4.6.5
Minimum sound pressure	3.26.14	4.6.3
Minimum of off-axis sound pressure	3.26.15	4.6.3
Amplifier endurance	3.26.16	4.6.6
Signalling circuits	3.26.17	4.6.8
Enclosure	3.3 and 3.9	MIL-STD-2036
Shock, vibration and inclination	3.3	MIL-STD-2036

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the equipment produced in one facility, using the same production processes and materials, and being offered for delivery at one time.

4.3.2 Sampling for group A examination and tests. As a minimum; the contractor shall randomly select a sample quantity of equipment in accordance with table II and subject them to the examinations and tests specified in 4.4. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100% of the lot for the defective characteristic(s) or providing a new lot which shall be examined and tested in accordance with the sampling plan contained herein.

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TABLE II. Sampling for group A examinations and tests.

Lot size	Sample size
2 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23

4.3.3 Sampling for group B tests. As a minimum, the contractor shall randomly select a sample quantity of equipment in accordance with table II and subject them to the examinations and tests specified in 4.4. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100% of the lot for the defective characteristic(s) or providing a new lot which shall be examined and tested in accordance with the sampling plan contained herein.

TABLE III. Sampling for group B tests.

Lot size	Sample size
2 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18

4.3.4 Sampling for group C tests. Group C tests will be required by the command or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to the group C tests listed in 4.4.

4.4 Quality conformance inspection. The sample equipments selected in accordance with 4.3 shall be subjected to the examination and tests specified in table IV (see 6.2). Tests shall be performed, in general, in the order listed.

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TABLE IV. Quality conformance inspection.

Inspection	Requirement	Test
Group A		
Examination	3.3	MIL-STD-2036
Operating	3.24	4.6.1
Amplifier gain	3.26.9	4.6.2
Group B		
Supply line voltage and frequency variation	3.3	MIL-STD-2036
Amplifier noise and microphonics	3.26.12	4.6.7
Overall frequency response	3.26.13	4.6.4 and 4.6.5
Minimum sound pressure	3.26.14	4.6.3
Minimum off-axis sound pressure	3.26.15	4.6.3
Signaling circuits	3.26.17	4.6.8
Group C		
Shielding and radio frequency noise reduction	3.3	MIL-STD-2036
Amplifier endurance	2.26.16	4.6.6
Temperature and humidity	3.3	MIL-STD-2036
Shock, vibration and inclination	3.3	MIL-STD-2036
Enclosure	3.3 and 3.9	MIL-STD-2036

4.5 Rejection. The results of the examination and each test shall be compared with specification requirements. Failure to conform to this specification for any group A, B or C examination and test shall be counted as a defect, and equipment shall not be offered for delivery.

4.6 Tests.

4.6.1 Operating test. The operating test shall consist of a voice transmission test for each intercommunication station with a complete operational test using a type LS-519/SIC or LS-518/SIC unit.

4.6.2 Amplifier gain. The sound source for the test shall be the test laboratory speech generator (artificial voice) adjusted so as to deliver a sound pressure of 4 dynes per cm^2 at a frequency of 1000 Hz under free field conditions at a point of axis and 457 mm in front of the ring on the speech generator. The loudspeaker to be tested shall be mounted in its front panel and placed 457 mm in front of and on axis of, and facing the ring on the speech generator. The loudspeaker shall be connected to a resistive load corresponding in value to the normal amplifier input impedance. The voltage developed across the resistor shall be measured to determine conformance with 3.26.9.

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4.6.3 Sound output pressure test. Three intercommunication stations shall be connected together for normal operation. One of the stations shall be placed with its reproducer at a point 457 mm in front of and on axis and facing the ring of the artificial voice. A sound source from the artificial voice shall be adjusted so as to deliver a sound pressure of 4 dynes per cm^2 at the warble band frequencies specified in 3.26.14 under free field conditions at a point on axis and 457 mm in front of the ring on the artificial voice. The station under test shall have the selector key for the two other stations depressed and the press-to-talk switch shall be operated. The sound output pressures at the two other stations (with volume control in maximum position) shall be measured. Both stations being transmitted to shall be acoustically insulated from each other.

4.6.4 Overall frequency response test using loudspeaker as a microphone. Conditions of overall frequency response test shall be as specified in 4.6.3 except that, in lieu of the warble frequencies specified in 3.26.14, a constant sound pressure at a frequency which shall be varied continuously from 200 to 8000 Hz shall be employed. The overall frequency response shall lie within the limits shown on figure 1.

4.6.5 Overall frequency response using a type M-132/SIC microphone. Test conditions shall be as specified in 4.6.4 except that the sound source pressure shall be 100 dynes per cm^2 and the distance from the microphone to the artificial mouth shall be 6.35 mm.

4.6.6 Amplifier endurance test. A resistive load of 1667 ohms shall be connected across the normal output terminals of a station. A 1000 Hz sine wave signal of such value as to produce rated output at rated output voltage shall be applied to the microphone input terminals. The station shall be energized and the press-to-talk switch shall be depressed for a period of 7 seconds and then released for a period of 15 seconds. Immediately upon completion of 20,000 operations the station shall be connected for normal operation. There shall be no noticeable change in the quality of speech transmitted, nor shall the output decrease greater than -2 dB from the output measured at the beginning of the test. Amplifier endurance tests shall be conducted in the hands-free mode, at rated load, for a period of 24 hours.

4.6.7 Amplifier noise and microphonics. With a 150-ohm resistor connected to the microphone input terminals, and full power applied to the amplifier, the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall be not greater than the following values when measured with an unweighted full range voltmeter. The measurement shall be made under two conditions of test, once while the component is held fixed and again while the component is under vibration in accordance with 3.26.12.

(a) Component stationary. Maximum measured voltage shall be 0.22 volt rms (50 dB below rated output voltage of 70 volts).
(b) Component under vibration. Maximum measured voltage shall be 0.31 volt rms (47 dB below rated output voltage of 70 volts).

4.6.8 Signaling circuits. A combination of three intercommunicating units shall be connected for normal operation. Cable used to connect the equipment shall be equivalent to the cables specified in 3.25.3. The equipment shall be tested in the following sequence:

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(a) At station number 1, select station numbers 2 and 3. The call lights at station numbers 2 and 3 should light. Repeat this test ten times.
(b) At station number 1, operate selector switch to select station number 2. At station number 3, operate selector switches to select stations numbers 1 and 2, consecutively. The busy light at station number 3 should light after each selection. This test shall be repeated ten times. A message shall be transmitted at station number 1 during the fifth and tenth test cycles. A listening test shall be made at station number 3 during the fifth and tenth test cycles. The transmission being made at station number 1 shall not be audible at station number 3.
(c) The test specified in 4.6.8(b) shall be repeated until each station in the system has served as the calling station (station number 1) and as the busy testing station (station number 3).
(d) Tests shall be conducted for compliance with 3.26.18 by operating station selector switches on both LS-458/SIC stations, simultaneously completing a call to a type LS-519/SIC or LS-518/SIC station.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Intercommunication stations.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the suppliers commercial practice. The information shall include nomenclature, federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

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5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged by level A; packed by level C, and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.3 and 6.2).

5.2.1 Levels of protection. The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking. The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified, and marked in accordance with MIL-E-17555.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushions etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiber board or transparent flexible cellular material.

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for intercommunication onboard Naval ships by means of voice amplification and station selection.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article testing is required (see 3.1).

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(d) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).
(e) When polystyrene (loose-fill) material is approved (see 5.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To insure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-NDTI-80809	Test reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manual must be acquired under separate contract line item in the contract. The manual shall be type I in accordance with MIL-M-15071.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.7 Subject term (key word) listing.

Amplifier-speaker
Master station
Signal circuit
Station selection
Voice amplification

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5830-N092)

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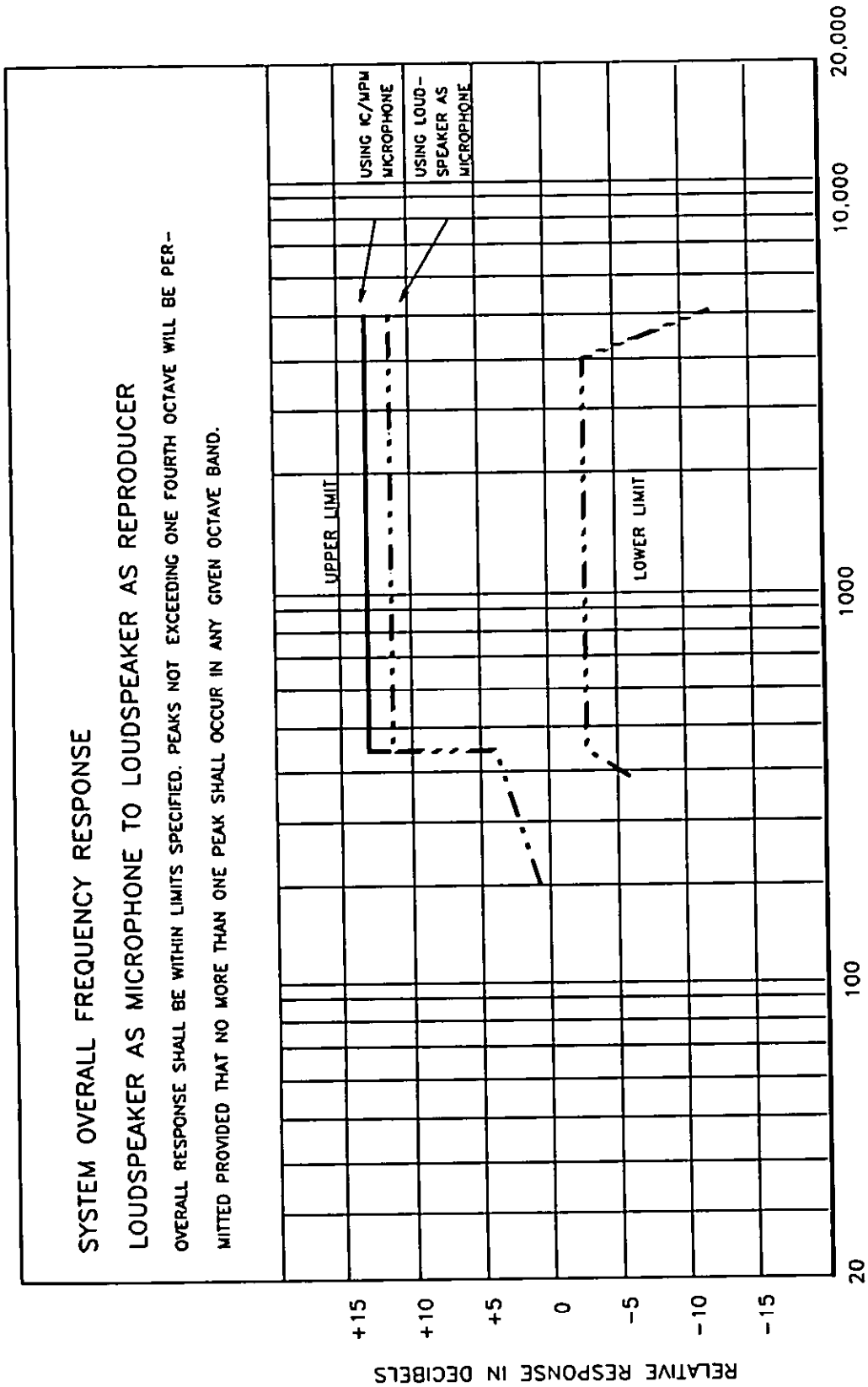


Figure 1 - System overall frequency response.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. *The submitter of this form must complete blocks 4, 5, 6, and 7.*

3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. *Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.*

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-22560C(SH)	2. DOCUMENT DATE (YYMMDD) 940426

3. DOCUMENT TITLE
INTERCOMMUNICATION STATIONS (HANDS-FREE OPERATED REPLY REPRODUCER) NAVAL SHIPBOARD

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

A. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

A. NAME Technical Point of Contact (TPOC) MR. MARK GOUMAS, SEA 03K32 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: 8-332-7191 TPOC: 703-602-7191 ext 317
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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